

2022

Penrith High School
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION



Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11–34 show relevant mathematical reasoning and/or calculations
- Write your NESA ID below, on the Multiple Choice Answer Sheet and the front of Booklets 1 & 2.

Total marks:
100

Section I – 10 marks (pages 3-8)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 91 marks (pages 9-32)

- Attempt Questions 11–34
- Allow about 2 hours and 45 minutes for this section

NESA NUMBER:

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TEACHER: _____

	Differentiation	Statistics	Algebra & Graphing	Integration	Series	Trigonometry
Multiple Choice:	/1	/2	/4	/1	/1	/1
	9	2, 6	3, 4, 7, 8	10	5	1

Differentiation	Statistics	Algebra & Graphing	Integration	Logarithms & Exponentials	Probability	Series	Trigonometry
/16	/10	/9	/19	/8	/10	/9	/20
Total:							/101

Section I

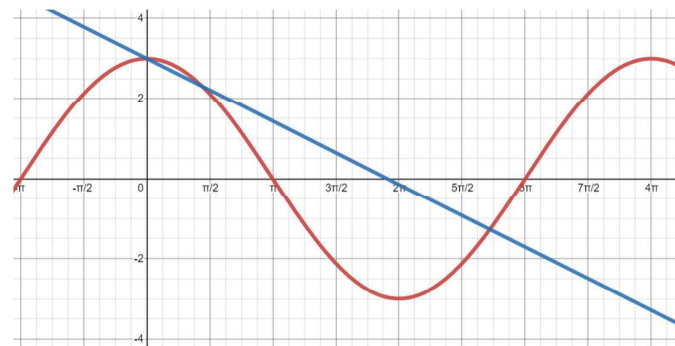
10 marks

Attempt questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

- 1 The graphs of $f(x) = 3 \cos\left(\frac{x}{2}\right)$ and $g(x) = 3 - \frac{1}{2}x$ are shown below.



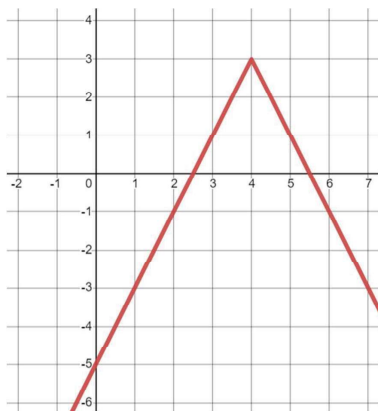
How many solutions are there to the equation $3 \cos\left(\frac{x}{2}\right) = -\frac{1}{2}x + 3$ for $0 < x < 4\pi$?

- A. 1
- B. 2
- C. 3
- D. 4

- 2 Mr Kim recorded the time it took him to drive to work over five days. The mean time was 37 minutes. The next workday it took Mr Kim 49 minutes to drive to work. What is his new mean time?

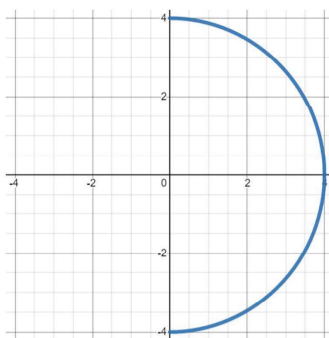
- A. 39
- B. 43
- C. 47
- D. 49

- 3 What is the equation of the graph shown below?



- A. $f(x) = -|x - 4| + 3$
 B. $f(x) = -2|x + 4| + 3$
 C. $f(x) = -|x + 4| + 3$
 D. $f(x) = -2|x - 4| + 3$

- 4 What type of relationship is represented by the graph shown below?



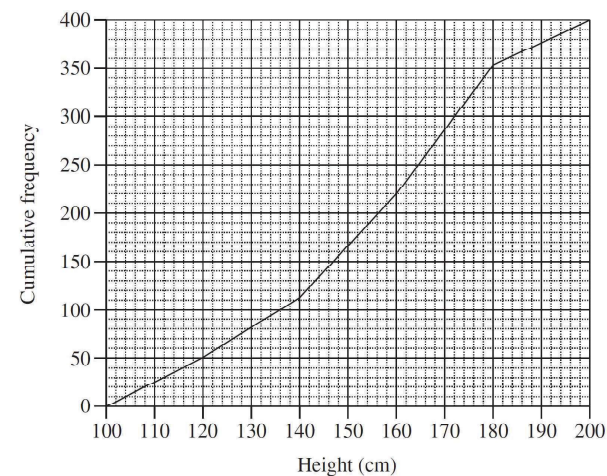
- A. Many-to-many
 B. Many-to-one
 C. One-to-many
 D. One-to-one

- 5 An investment of \$7 000 gains 3% per annum for 20 years, with interest calculated monthly.

Which expression gives the value of the investment after 20 years?

- A. $7\,000(1 + 0.03)^{20}$
 B. $7\,000\left(1 + \frac{0.03}{12}\right)^{240}$
 C. $7\,000(1 + (0.03 \times 12))^{\frac{20}{12}}$
 D. $7000 \times 0.03 \times 20$

- 6 The height of 400 students were measured. The results are displayed in the cumulative frequency polygon shown below.



What is the median height of the students?

- A. 150
 B. 156
 C. 165
 D. 200

7 Let $f(x) = \frac{2}{3-x}$ and $g(x) = x - 5$. The domain of $f(g(x))$ is

- A. All real x
- B. $x \neq 2$
- C. $x \neq 3$
- D. $x \neq 8$

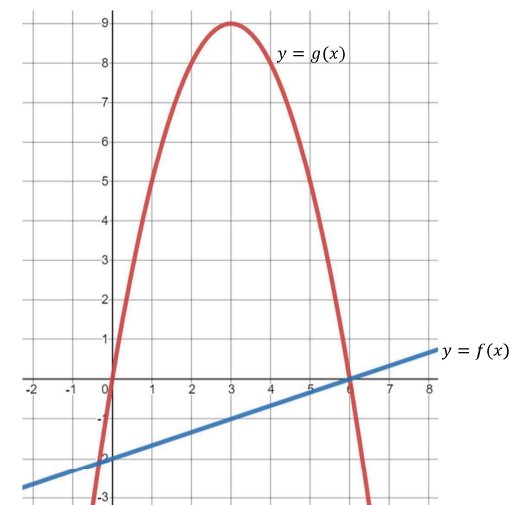
8 The radius and centre of a circle with equation $x^2 + y^2 - 8y - 1 = 0$ is

- A. Radius = 17, Centre (0, -4)
- B. Radius = 15, Centre (0, 4)
- C. Radius = $\sqrt{15}$, Centre (0, -4)
- D. Radius = $\sqrt{17}$, Centre (0, 4)

9 Find the derivative of $e^{x \cos 3x}$.

- A. $e^{3x \sin 3x}$
- B. $e^{x \cos 3x}$
- C. $e^{x \cos 3x}(\cos 3x + 3x \sin 3x)$
- D. $e^{x \cos 3x}(\cos 3x - 3x \sin 3x)$

10 The graphs of $y = g(x)$ and $y = f(x)$ are shown below.



It is known that $\int_0^6 (g(x) - f(x)) dx = 42$.

Find $\int_0^6 g(x) dx$

- A. 30
- B. 31.5
- C. 36
- D. 38

Question 11 (3 marks)

Find the equation of the tangent to the curve $y = x^3 - 4x^2$ at the point $(2, -8)$.

[illegible]

Question 12 (2 marks)

Solve $5 - 3x < 20$, giving your answer in interval notation.

[illegible]

Question 13 (3 marks)

The function $f(x) = e^x$ is transformed to $g(x) = -e^{3(x+4)}$.

Describe, in words, the sequence of transformations.

Question 14 (2 marks)

Differentiate $\frac{x^3 + 2}{x - 1}$.

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B

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Question 15 (2 marks)

What is the limiting sum of the following geometric series?

$$-324 + 108 - 36 + 12 \dots$$

[illegible]

Solve $2 \log_2 x - \log_2(x + 6) = 3$.

[illegible]

Evaluate $\int_0^{\frac{\pi}{2}} \cos\left(\frac{x}{3}\right) dx$.

This image shows a full page of primary-ruled notebook paper. It features ten sets of horizontal lines across the page. Each set consists of a solid top blue line, a dashed middle blue line, and a solid bottom blue line, providing a guide for letter height and placement. The paper is otherwise blank, with no handwriting or other markings.

4

Mr Tan played a table tennis match. The number of points, X , that he scores in each set is a random variable with probability distribution given by:

(a) Evaluate a and b given that Mr Tan has an expected score of 10.4 per set.

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[illegible]

3

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Question 19 (6 marks)

A semi circle with radius 5 has equation $f(x) = \sqrt{25 - x^2}$.

- (a) Use the trapezoidal rule with five function values to approximate the area of the semi circle. Answer correct to 3 decimal places.

3

- (b) Calculate the difference between the actual area of the semi circle and the approximation by the trapezoidal rule. Answer correct to 3 decimal places.

2

- (c) Explain why the trapezoidal rule gives an estimate that is lower than the actual area of the semi circle.

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Question 20 (3 marks)

Mr Huynh recorded how many push ups he did each day for thirty days.

3

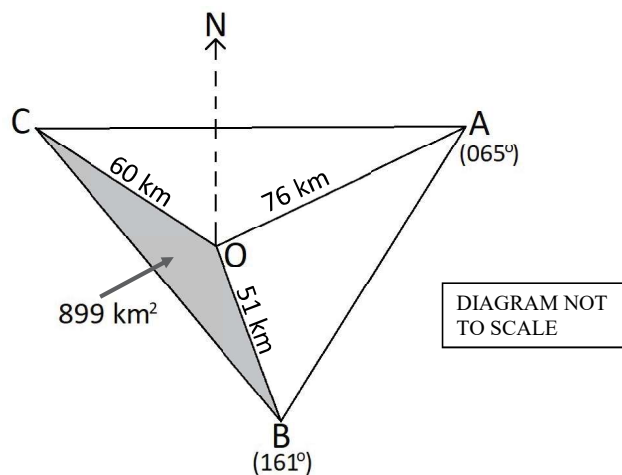
The results are shown in the frequency table.

Push ups per day	Frequency
100	10
120	6
140	12
160	1
180	0
200	0
220	1

Using calculations, decide whether or not 220 push ups was an outlier.

Question 21 (4 marks)

The diagram shows three triangular blocks of land which all share a common point O . The distance and true bearings of the corners of the blocks from point O are shown.



The area of the obtuse-angled triangle BOC is 899 km^2 .

Calculate the true bearing of the point C from the point O , correct to the nearest degree.

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Question 22 (3 marks)

Mrs Briggs notices that the amount of emails she receives per week from students is cyclical. Every 12 weeks she receives the highest number of emails. The average amount of emails per week is 25 and the highest number of emails per week is 40.

The number of emails Mrs Briggs receives each week, $f(t)$, is given by

$$f(t) = a \cos(bt) + c$$

where t is the number of weeks after 1 January 2022 and $0 \leq t \leq 52$.

The number of emails is at a maximum when $t = 0$.

Find the values a, b and c .

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Question 23 (4 marks)

Currently, there is a mouse plague in Western NSW. The population of mice can be found using the formula $N = 10(e^{kt})$, where N is the number of mice and t is the time in months.

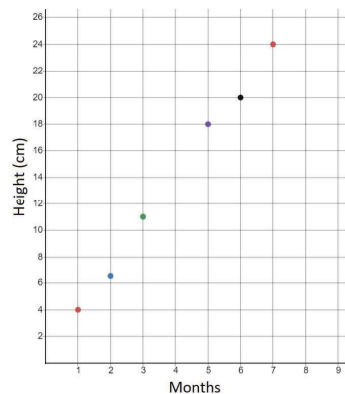
The plague started with 10 mice and increased to 100 mice after 3 months.

By first finding the value of k , show that the exact rate of increase after 12 months is $\frac{10\ln 10}{3}(e^{100000})$ mice/month.

[illegible]

The collected data is shown in the table and graphed in the scatterplot shown.

Months since planting, m .	1	2	3	4	5	6	7
Height of plant, h cm	4	6.5	11	x	18	20	24



- (b) Calculate Pearson's correlation coefficient for the data, correct to three decimal places. 1

- 20 -

(d) Discuss the accuracy of your answer to part (c). 1

- (e) Interpret the value of the gradient of the least squares regression line in the given context. 1

Prove that $\tan\theta\sin\theta + \cos\theta = \sec\theta$. 3

- 21 -

Question 26 (5 marks)

Sketch the graph of the curve $y = x^3 - 3x^2 + 1$, labelling the stationary points and point of inflection. 5
Do NOT determine the x -intercepts of the curve.

[illegible]

Question 27 (2 marks)

Mr Pollard is creating a website. Every weekend, apart from the first, he adds the same number of pages to his website. After four weekends he has added 25 pages and after eleven weekends he has added 74 pages.

- (a) Show that the total number amount of pages, P , added after the n th weekend can be modelled by the recurrence relation

$$P_n = P_{n-1} + 7, \text{ where } n = 2, 3, 4 \dots$$

- (b) Given that on the first weekend he only added 4 pages, use the recurrence relation to find the total number of pages Mr Pollard has added after the third weekend.

Question 28 (4 marks)

Dr Katyal pours 500 mL of liquid fertiliser into a previously empty bucket. She then fills the bucket with water at a rate of $\frac{dV}{dt}$ litres per minute, where $\frac{dV}{dt} = \frac{5}{t+1}$ and t is time in minutes. Find the time, to the nearest second, at which there is 10 L of liquid in the bucket.

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Question 29 (7 marks)

Each year, the number of ice creams sold, $c(t)$, can be modelled by the function

$$c(t) = 200 \cos\left(\frac{\pi}{26}t\right) + 300$$

where t is the number of weeks after January 1st.

- (a) Find how many ice creams were sold on January 1st.

1

- (b) What are the two values of t for which there were 400 ice creams sold?

3

Question 29 continues on page 23

Question 29 (continued)

- (b) Sketch the graph of $c(t)$ for $0 \leq t \leq 52$.

3

Question 30 (3 marks)

- (a) Show that the derivative of $2x \log_5 x$ is $2 \log_5 x + \frac{2x}{x \ln 5}$

2

- (b) Hence, find $\int_0^{\frac{\pi}{2}} \log_5 x + \frac{1}{\ln 5} dx$.

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Question 31 (4 marks)

Ms Alrubai and Mrs Norman are employed as doctors. They both start on the same annual salary. However, Ms Alrubai negotiated an annual salary increase of \$5 000, while Mrs Norman negotiated an annual salary increase of 7%.

4

Find their starting annual salary given that after 16 years they have both made the same amount of money in total.

Question 32 (6 marks)

If Mr Ferrarin wakes up late, the probability that he exercises is $\frac{1}{3}$, and if he wakes up on time the probability that he exercises is $\frac{4}{5}$.

- (a) In a particular week, Mr Ferrarin wakes up on time on Thursday morning and late on Friday morning.

2

Find the probability that Mr Ferrarin exercised on at least one of these days.

- (b) The probability that Mr Ferrarin wakes up on time on a Monday is $\frac{1}{4}$.

- (i) Find the probability that Mr Ferrarin exercises on any given Monday.

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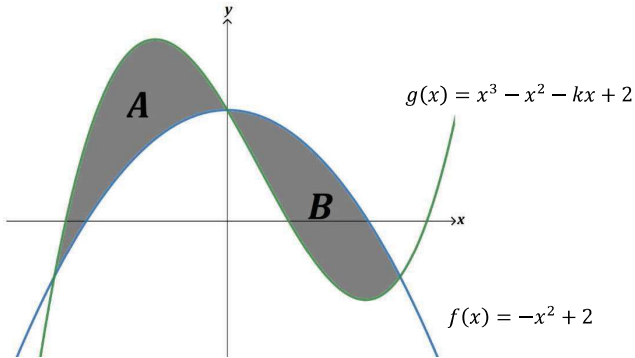
- (ii) Hence, or otherwise, find the probability that Mr Ferrarin woke up on time on a Monday, given that he exercised that day.

2

Question 33 (4 marks)

The graphs of $f(x) = -x^2 + 2$ and $g(x) = x^3 - x^2 - kx + 2$, where k is a constant, are shown below.

The graphs intersect and create two closed regions, A and B.



Show that these two regions have the same area.

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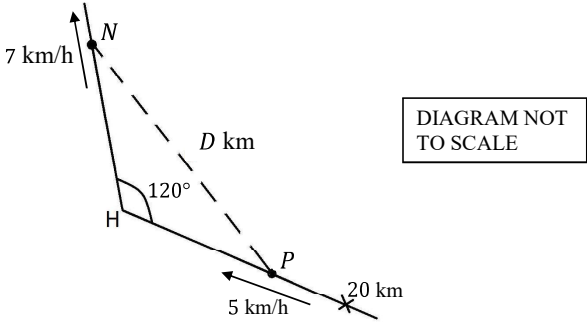
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Question 34 (5 marks)

Ned Kelly, N , left his hideout, H , and walked along a straight path at 7 km/h. At the same time that Ned left his hideout, the Police, P , are 20 km away from his hideout, walking towards it along a different straight path at 5 km/h. The two paths meet at an angle of 120° .



The distance between Ned Kelly and the police at time t hours is D km.

- (a) Show that $D^2 = 39t^2 - 60t + 400$.

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- (b) Find the minimum distance, to the nearest kilometre, between the police and Ned Kelly.

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Mathematics Advanced

Total marks:
100

Section I – 10 marks (pages 2-5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6–30)

- Attempt Questions 11–34
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NESA NUMBER:

SOLUTIONS

TEACHER:

	Differentiation	Statistics	Algebra & Graphing	Integration	Series	Trigonometry
Multiple Choice:	9 /1	2, 6 /2	3, 4, 7, 8 /4	10 /1	5 /1	1 /1

Differentiation	Statistics	Algebra & Graphing	Integration	Logarithms & Exponentials	Probability	Series	Trigonometry
/20	/10	/9	/15	/8	/10	/9	/20
Total:							/101

Section I

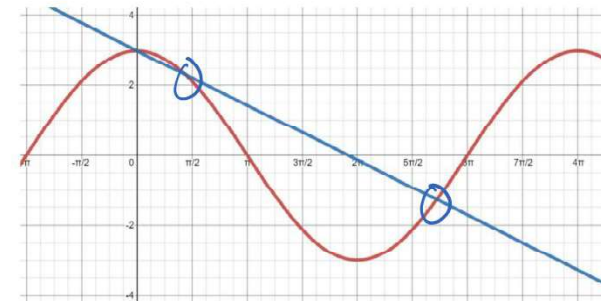
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How many solutions are there to the equation $3 \cos\left(\frac{x}{2}\right) = -\frac{1}{2}x + 3$ for $0 < x < 4\pi$?

- A. 1
B. 2
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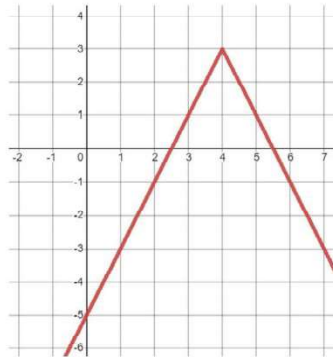
don't include $x=0$

- 2 Mr Kim recorded the time it took him to drive to work over five days. The mean time was 37 minutes. The next workday it took Mr Kim 49 minutes to drive to work. What is his new mean time?

- A. 39
B. 43
C. 47
D. 49

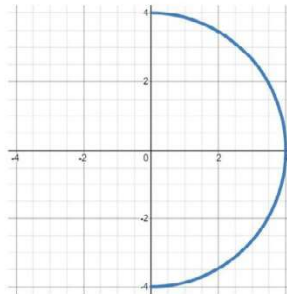
$$\frac{37 \times 5 + 49}{6} = 39$$

- 3 What is the equation of the graph shown below?



- A. $f(x) = -|x - 4| + 3$
 B. $f(x) = -2|x + 4| + 3$
 C. $f(x) = -|x + 4| + 3$
 D. $f(x) = -2|x - 4| + 3$

- 4 What type of relationship is represented by the graph shown below?

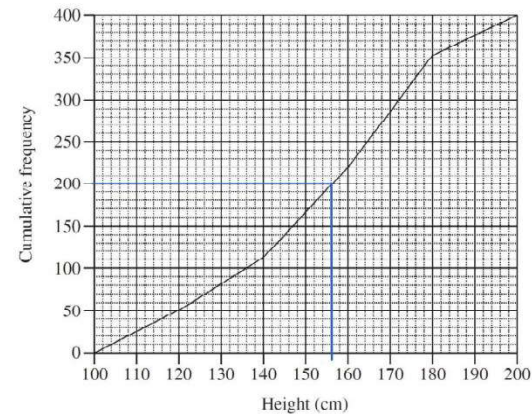


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 B. Many-to-one
 C. One-to-many
 D. One-to-one

- 5 An investment of \$7 000 gains $\frac{0.03}{12}$ per annum for 20 years, with interest calculated monthly. $20 \times 12 = 240$
 Which expression gives the value of the investment after 20 years?

- A. $7\,000(1 + 0.03)^{20}$
 B. $7\,000\left(1 + \frac{0.03}{12}\right)^{240}$
 C. $7\,000(1 + (0.03 \times 12))^{\frac{20}{12}}$
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- 6 The height of 400 students were measured. The results are displayed in the cumulative frequency polygon shown below.



What is the median height of the students?

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 B. 156
 C. 165
 D. 200

7 Let $f(x) = \frac{2}{3-x}$ and $g(x) = x - 5$. The domain of $f(g(x))$ is

- A. All real x
- B. $x \neq 2$
- C. $x \neq 3$
- D. $x \neq 8$

$$\frac{2}{3-(x-5)} \\ = \frac{2}{3-x+5} \quad \therefore -x+8 \neq 0 \\ = \frac{2}{-x+8} \quad x \neq 8$$

8 The radius and centre of a circle with equation $x^2 + y^2 - 8y - 1 = 0$ is

- A. Radius = 17, Centre (0, -4)
- B. Radius = 15, Centre (0, 4)
- C. Radius = $\sqrt{15}$, Centre (0, -4)
- D. Radius = $\sqrt{17}$, Centre (0, 4)

$$x^2 + y^2 - 8y + \left(\frac{8}{2}\right)^2 = \left(\frac{8}{2}\right)^2 + 1 \\ x^2 + (y-4)^2 = 17$$

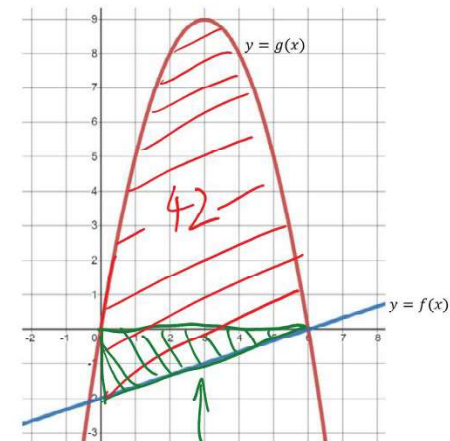
9 Find the derivative of $e^{x \cos 3x}$.

- A. $e^{3x \sin 3x}$
- B. $e^{x \cos 3x}$
- C. $e^{x \cos 3x} (\cos 3x + 3x \sin 3x)$
- D. $e^{x \cos 3x} (\cos 3x - 3x \sin 3x)$

$$y = e^{f(x)} \\ y' = f'(x) e^{f(x)}$$

$$u = x \\ u' = 1 \\ v = \cos 3x \\ v' = -3 \sin 3x \\ f'(x) = \cos 3x - 3x \sin 3x$$

10 The graphs of $y = g(x)$ and $y = f(x)$ are shown below.



It is known that $\int_0^6 (g(x) - f(x)) dx = 42$.

Find $\int_0^6 g(x) dx$.

- A. 30
- B. 31.5
- C. 36
- D. 38

$$A = \frac{1}{2} \times 6 \times 2 \\ = 6$$

$$\int_0^6 g(x) dx = 42 - 6 \\ = 36$$

Question 11 (3 marks)

Find the equation of the tangent to the curve $y = x^3 - 4x^2$ at the point $(2, -8)$.

$$y - y_1 = m(x - x_1) \quad y' = 3x^2 - 8x$$

$$y - (-8) = -4(x - 2) \quad \text{at } x = 2$$

$$y + 8 = -4x + 8 \quad m = 3(2)^2 - 8(2)$$

$$y = -4x \quad = -4$$

Question 12 (2 marks)

Solve $5 - 3x < 20$, giving your answer in interval notation.

$$-3x < 15$$

$$x > -5$$

$$(-5, \infty)$$

Question 13 (3 marks)

The function $f(x) = e^x$ is transformed to $g(x) = -e^{3(x+4)} = -e^{3x+12}$

Describe, in words, the sequence of transformations.

Reflection across the x -axis
Horizontal dilation by a factor of 3
Horizontal translation, left 4 units

Note: The reflection can be placed anywhere, as long as the horizontal transformations are in the correct order.

Question 14 (2 marks)

Differentiate $\frac{x^3 + 2}{x - 1}$

$$u = x^3 + 2 \quad v = x - 1$$

$$u' = 3x^2 \quad v' = 1$$

$$\frac{(3x^2)(x-1) - (x^3+2)(1)}{(x-1)^2}$$

$$= \frac{3x^3 - 3x^2 - x^3 - 2}{(x-1)^2} = \frac{2x^3 - 3x^2 - 2}{(x-1)^2}$$

Question 15 (2 marks)

What is the limiting sum of the following geometric series?

$$-324 + 108 - 36 + 12 \dots$$

$$S = \frac{a}{1-r}$$

$$a = -324$$

$$r = \frac{108}{-324} = -\frac{1}{3}$$

$$= \frac{-324}{1 - (-\frac{1}{3})} = -243$$

Question 16 (4 marks)

Solve $2\log_2 x - \log_2(x+6) = 3$.

$$\log_2 x^2 - \log_2(x+6) = 3$$

Students need to revise the log laws.

$$\log_2 \frac{x^2}{x+6} = 3$$

$$\frac{x^2}{x+6} = 2^3$$

$$x^2 = 8(x+6)$$

$$x^2 = 8x + 48$$

$$x^2 - 8x - 48 = 0$$

$$(x-12)(x+4) = 0$$

$$x = -1 \pm \sqrt{57}$$

2
52mks

$$x = 12 \text{ or } -4$$

$$x = 12 \quad (x \geq 0)$$

most students forgot the last step.

Question 17 (3 marks)

Evaluate $\int_0^{\frac{\pi}{2}} \cos\left(\frac{x}{3}\right) dx$.

$$= \int_0^{\frac{\pi}{2}} \cos\left(\frac{1}{3}x\right) dx$$

$$f(x) = \frac{1}{3}x$$

$$f'(x) = \frac{1}{3}$$

$$= 3 \int_0^{\frac{\pi}{2}} \frac{1}{3} \cos\left(\frac{1}{3}x\right) dx$$

$$= 3 \left[\sin\left(\frac{1}{3}x\right) \right]_0^{\frac{\pi}{2}}$$

$$3\left(\frac{1}{2} - 0\right)$$

$$= 3 \left[\sin\left(\frac{1}{3} \times \frac{\pi}{2}\right) - \sin\left(\frac{1}{3} \times 0\right) \right]$$

$$= \frac{3}{2}$$

$$= 3 \left[\sin\left(\frac{\pi}{6}\right) - \sin 0 \right]$$

$$I = \left[3 \sin \frac{x}{3} \right]_0^{\frac{\pi}{2}}$$

Question 18 (4 marks)

Mr Tan played a table tennis match. The number of points, X , that he scores in each set is a random variable with probability distribution given by:

$X = x$	9	10	11	12
$P(x)$	0.2	a	b	0.1

(a) Evaluate a and b given that Mr Tan has an expected score of 10.4 per set.

$$0.2 + a + b + 0.1 = 1$$

$$a + b = 0.7$$

$$a = 0.7 - b$$

$$9 \times 0.2 + 10a + 11b + 12 \times 0.1 = 10.4$$

$$1.8 + 10a + 11b + 1.2 = 10.4$$

$$10a + 11b = 7.4$$

$$\therefore 10(0.7 - b) + 11b = 7.4$$

$$7 - 0b + 11b = 7.4$$

$$b = 0.4$$

$$a = 0.7 - 0.4 = 0.3$$

$$a = 0.3, b = 0.4$$

If you just had $a = 0.3$ and $b = 0.4$ with no working $\therefore$ max. 1 mark.

(b) Calculate the variance of X .

x^2	81	100	121	144
-------	----	-----	-----	-----

$$E(x^2) = 81 \times 0.2 + 100 \times 0.3 + 121 \times 0.4 + 144 \times 0.1 = 109$$

$$E(x^2) - \mu^2 = 109 - 10.4^2 = 109 - 108.16 = 0.84$$

right or wrong.

Question 19 (6 marks)

A semi-circle with radius 5 has equation $f(x) = \sqrt{25 - x^2}$.

means 4 intervals.



- (a) Use trapezoidal rule with five function values to approximate the area of the semi-circle.

Answer correct to 3 decimal places.

① $A = \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + f(x_2) + f(x_3)]\}$

$a = -5$ $f(-5) = 0$
 $x_1 = -2.5$ $f(-2.5) = \sqrt{18.75}$
 $x_2 = 0$ $f(0) = 5$
 $x_3 = 2.5$ $f(2.5) = \sqrt{18.75}$
 $b = 5$ $f(5) = 0$

$= \frac{5 - (-5)}{2 \times 4} \{0 + 0 + 2[\sqrt{18.75} + 5 + \sqrt{18.75}]\}$

$= 34.15063509$

$= 34.151 \text{ units}^2$ (3dp)

* very badly done.
 * You must write out the formula and line of substitution

- (b) Calculate the difference between the actual area of the semi-circle and the approximation by trapezoidal rule. Answer correct to 3 decimal places.

$A = \frac{1}{2} \pi \times 5^2$

$= \frac{25}{2} \pi$

$= 39.2699$

$\text{Diff} = \frac{25}{2} \pi - 34.151$

$= 5.119 \text{ units}^2$ less by trapezoidal rule.

- (c) Explain why trapezoidal rule gives an estimate that is lower than the actual area of the semi-circle.

Trapezoidal rule gives a lower estimate because when you draw the trapeziums on a concave down graph, they will always be below the curve, as shown



in the diagram.

Question 20 (3 marks)

Mr Huynh recorded how many push ups he did each day for thirty days.

The results are shown in the frequency table.

Push ups per day	Frequency	c.f.
100	10	10
120	6	16
140	12	28
160	1	29
180	0	29
200	0	29
220	1	30

Using calculations, decide whether or not 220 push ups was an outlier.

Median = 15.5th score = 120

$Q_1 = 8\text{th score} = 100$

$Q_3 = 23\text{rd score} = 140$

$QR = 140 - 100 = 40$

$Q_3 + 1.5 \times QR = 140 + 1.5 \times 40$

$= 160$

$220 > 160$

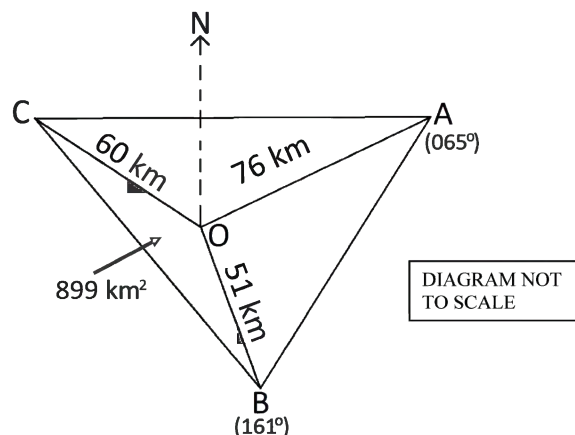
$\therefore 220$ is an outlier.

OC

Answer only = 1 mark.

Question 21 (4 marks)

The diagram shows three triangular blocks of land which all share a common point O . The distance and true bearings of the corners of the blocks from point O are shown.



The area of the obtuse-angled triangle BOC is 899 km^2 .

Calculate the true bearing of the point C from the point O , correct to the nearest degree.

$$\begin{aligned} \text{Bearing} &= 161 + \angle BOC \\ &= 161 + 144 \\ &= 305^\circ \quad \text{CFM} \end{aligned}$$

$$\begin{aligned} A &= \frac{1}{2}ab \sin C \\ 899 &= \frac{1}{2} \times 51 \times 60 \times \sin(\angle BOC) \\ \frac{899}{1530} &= \sin(\angle BOC) \\ \angle BOC &= \sin^{-1}\left(\frac{899}{1530}\right) \\ &= 35.99 \\ &= 36^\circ \end{aligned}$$

Many students ignored the "obtuse-angled" part of the question or didn't know how to find the obtuse angle (adding 90° was a common error).

Some ignored the area of the triangle part of the question. Instead using cosine or sine rule to find irrelevant sides or angles.

But it is an obtuse angled triangle.

$$\therefore 180 - 36 = 144$$

There are no tricks in the HSC. If the information is there then use it! Also use the reference sheet!

Question 22 (3 marks)

Mrs Briggs notices that the amount of emails she receives per week from students is cyclical. Every 12 weeks she receives the highest number of emails. The average amount of emails per week is 25 and the highest number of emails per week is 40.

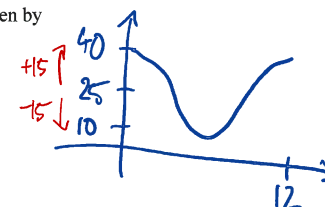
The number of emails Mrs Briggs receives each week, $f(t)$, is given by

$$f(t) = a \cos(bt) + c$$

Where t is time in weeks and $0 \leq t \leq 52$.

The number of emails is at a maximum when $t = 0$.

Find the values a , b and c .



$$\begin{aligned} \text{period} &= \frac{2\pi}{b} \\ 12 &= \frac{2\pi}{b} \\ b &= \frac{\pi}{6} \end{aligned}$$

$$\begin{aligned} a &= 15 \\ c &= 25 \end{aligned}$$

Draw a picture! It's easier.

After feedback from last exam, many students learnt the period formula. However some have still not.

Question 23 (4 marks)

Currently, there is a mouse plague in Western NSW. The population of mice can be found using the formula $N = 10(e^{kt})$, where N is the number of mice and t is the time in months.

The plague started with 10 mice and increased to 100 mice after 3 months.

Rate = derivative.

By first finding the value of k , show that the exact rate of increase after 12 months is $\frac{10 \ln 10}{3} (e^{\ln 10000})$ mice/month.

$$\begin{aligned} 100 &= 10(e^{3k}) \\ 10 &= e^{3k} \\ \ln 10 &= 3k \\ k &= \frac{\ln 10}{3} \end{aligned}$$

$$N' = 10 \times \frac{\ln 10}{3} e^{\frac{\ln 10}{3} t}$$

at $t = 12$

$$N' = \frac{10 \ln 10}{3} e^{\frac{\ln 10}{3} \times 12}$$

$$= \frac{10 \ln 10}{3} e^{4 \ln 10}$$

$$= \frac{10 \ln 10}{3} e^{\ln 10^4}$$

$$= \frac{10 \ln 10}{3} e^{\ln 10000}$$

Always leave as exact value. Never decimals, unless the Q asks.

This was poorly done. It is a very common exam question. If you didn't get it then please practice this type of question.

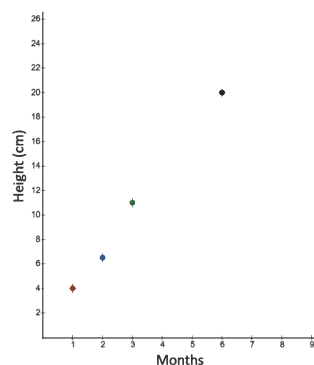
Must have both of these lines for final mark. This is a show Q. You can't jump from $4 \ln 10$ to $\ln 10000$

Question 24 (5 marks)

Mrs Zhou studied the growth of one of her indoor plants over several months.

The collected data is shown in the table and graphed in the scatterplot shown.

Months since planting, m .	1	2	3	4	5	6	7
Height of plant, h cm	4	6.5	11	x	18	20	24



- (a) Determine the equation of the least-squares regression line for this data.

1

$$A = 0.488 \quad B = 3.357$$

$$h = 3.357m + 0.488$$

- (b) Calculate Pearson's correlation coefficient for the data, correct to three decimal places.

1

$$r = 0.997$$

Overall, all of Q24 was well done. Again a big improvement from last exam. Again these are very common (and very easy) questions. If you still didn't get it then practice these types of questions!

Question 24 continues on page 19

Question 24 (Continued)

- (c) Mrs Zhou missed recording the plants' height at four months. Use the least squares regression line to estimate the height, x , of the plant at four months. Answer correct to one decimal place.

1

$$h = 3.357(4) + 0.488$$

$$= 13.9 \text{ cm (1dp)}$$

Show the substitution! Otherwise you may not get a CFM if your part a was wrong.

- (d) Discuss on the reliability of your answer to part (c).

1

The estimate is very reliable, as Pearson's correlation coefficient is 0.997 and it is interpolation.

Interpolation on its own does not mean a prediction is reliable, it needs to be combined with a Pearson's Correlation Coefficient close to 1 or -1. Just as extrapolation on its own does not mean a prediction is unreliable.

- (e) Interpret the value of the gradient of the least squares regression line in the given context.

1

The gradient means the plant grows, on average, 3.357 cm per month.

Key words in the question are Interpret, value of the gradient, context. You must mention the actual value (3.357) and what it means (growth in cm/month).

Question 25 (3 marks)

Prove that $\tan\theta\sin\theta + \cos\theta = \sec\theta$.

3

$$\text{LHS} = \tan\theta\sin\theta + \cos\theta$$

$$\begin{aligned} &= \frac{\sin\theta}{\cos\theta} \sin\theta + \cos\theta \\ &= \frac{\sin^2\theta}{\cos\theta} + \cos\theta \\ &= \frac{\sin^2\theta}{\cos\theta} + \frac{\cos^2\theta}{\cos\theta} \\ &= \frac{\sin^2\theta + \cos^2\theta}{\cos\theta} \end{aligned}$$

Again this feedback was given in a past exam and still some didn't learn it.

This is a show question. Don't skip steps! Many skipped the first line. Lucky to get the mark.

$$\begin{aligned} &= \frac{1}{\cos\theta} \\ &= \sec\theta \\ &= \text{RHS} \end{aligned}$$

BOTH ①

Question 26 (5 marks)

Sketch the graph of the curve $y = x^3 - 3x^2 + 1$, labelling the stationary points and point of inflection. Do NOT determine the x -intercepts of the curve.

The verb is sketch. Your graph is the important bit. Take your time and don't make it small. Use another page if needed.

Label coordinates for 1 mark.

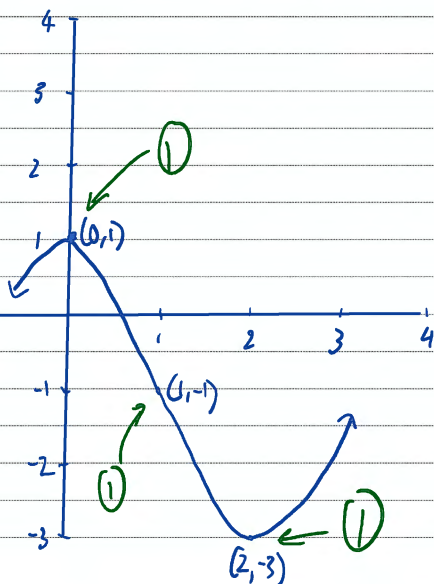
$y' = 3x^2 - 6x = 0$
 $3x(x-2) = 0$
 $x = 0 \text{ or } 2$
 $y = (0)^3 - 3(0)^2 + 1 = 1$
 $y = (2)^3 - 3(2)^2 + 1 = -3$
 Stationary points: $(0, 1)$ and $(2, -3)$

possible pt of inflection when $y'' = 0$
 $y'' = 6x - 6 = 0$
 $6x = 6$
 $x = 1$
 $y = (1)^3 - 3(1)^2 + 1 = -1$
 Point of inflection: $(1, -1)$

TEST: $x \quad 0.9 \quad 1 \quad 1.1$
 $y'' \quad -0.6 \quad 0 \quad 0.6$
 $\therefore (1, -1)$ is pt of inflection
 $(0, 1)$ is max
 $(2, -3)$ is min

Many students don't test the point of inflection.

Some students lost a mark for really bad scales. There are lines on the page, use them for your scale! They also make bad scales obvious.



Question 27 (2 marks)

Mr Pollard is creating a website. Every weekend, apart from the first, he adds the same number of pages to his website. After four weekends he has added 25 pages and after eleven weekends he has added 74 pages.

- (a) Show that the total number amount of pages, P , added after the n th weekend can be modelled by the recurrence relation

$$P_n = P_{n-1} + 7, \text{ where } n = 2, 3, 4, \dots$$

$$\begin{aligned}
 P_4 + 7d &= P_{11} \\
 25 + 7d &= 74 \\
 7d &= 49 \\
 d &= 7
 \end{aligned}$$

- (b) Given that on the first weekend he only added 4 pages, use the recurrence relation to find the total number of pages Mr Pollard has added after the third weekend.

Must have this! (use the recurrence relation.)
 $P_0 = 4$
 $P_1 = 4 + 7 = 11$
 $P_2 = 11 + 7 = 18$
 $\therefore$ He has 18 pages after the third weekend.

Question 28 (4 marks)

Dr Katyal pours 500 mL of liquid fertiliser into a previously empty bucket. She then fills the bucket with water at a rate of $\frac{dV}{dt}$ litres per minute, where $\frac{dV}{dt} = \frac{5}{t+1}$ and t is time in minutes. Find the time, to the nearest second, at which there is 10 L of liquid in the bucket.

$\int \frac{dV}{dt} dt = V$
 $V = \int \frac{5}{t+1} dt$
 $= 5 \ln|t+1| + C$
 when $t=0$, $V=0.5$
 $0.5 = 5 \ln|0+1| + C$
 $C = 0.5$

$\int \frac{f'(x)}{f(x)} = \ln|f(x)| + C$
 $f(x) = t+1$
 $f'(x) = 1$
 This formula is on the reference sheet!

$V = 5 \ln|t+1| + 0.5$
 when $V=10$
 $10 = 5 \ln|t+1| + 0.5$
 $\frac{9.5}{5} = \ln|t+1|$
 $t+1 = e^{\frac{9.5}{5}}$
 $t = e^{\frac{9.5}{5}} - 1$
 $= 5.686$
 $= 5 \text{ min } 41 \text{ sec.}$

This is a very common exam question. If you got it wrong then practice!

Some students wrote 6 sec, but t is in minutes.

Question 29 (7 marks)

Each year, the number of ice creams sold, $c(t)$, can be modelled by the function

$$c(t) = 200 \cos\left(\frac{\pi}{26}t\right) + 300$$

where t is the number of weeks after January 1st.

- (a) Find how many ice creams were sold on January 1st.

1

$$c(0) = 200 \cos\left(\frac{\pi}{26} \times 0\right) + 300$$

$$= 500 \quad (1)$$

- (b) What are the two values of t for which there were 400 ice creams sold?

3

$$400 = 200 \cos\left(\frac{\pi}{26}t\right) + 300$$

$$100 = 200 \cos\left(\frac{\pi}{26}t\right)$$

$$\frac{1}{2} = \cos\left(\frac{\pi}{26}t\right) \quad (1)$$

$$\text{Ref } L = \frac{\pi}{3} \quad \frac{\text{S/A}}{\text{T/C}}$$

$$\frac{\pi}{26}t = \frac{\pi}{3}, 2\pi - \frac{\pi}{3} \quad (1)$$

$$= \frac{\pi}{3}, \frac{5\pi}{3}$$

$$t = \frac{26}{3}, \frac{130}{3} \quad (1)$$

* Most students did it well.

* If only one solution 2/3

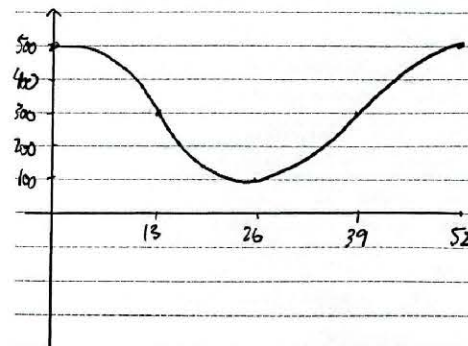
* If degrees then final answer incorrect
↳ unless converted to radians again.

Question 29 (continued)

- (b) Sketch the graph of $c(t)$ for $0 \leq t \leq 52$.

3

$$\begin{aligned} \text{period} &= \frac{2\pi}{a} \\ &= \frac{2\pi}{\frac{\pi}{26}} \\ &= 52 \end{aligned}$$



(i) shape

(i) Amplitude from 500 to 100

(i) Period (and correctly labelled axis)

* If arrows then incorrect

* If not symmetrical then -1.

Question 30 (3 marks)

- (a) Show that the derivative of $2x \log_5 x$ is $2 \log_5 x + \frac{2x}{x \ln 5}$

Need to show this

$$\begin{aligned} & \text{① } \left[\begin{array}{l} u = 2x \\ u' = 2 \\ v = \log_5 x \\ v' = \frac{1}{(\ln 5) x} = \frac{1}{x \ln 5} \end{array} \right] \text{①} \end{aligned}$$

** Need to show line of substitution otherwise copying from answers.*

$$2 \log_5 x + 2x \left(\frac{1}{x \ln 5} \right) = 2 \log_5 x + \frac{2x}{x \ln 5}$$

- (b) Hence, find $\int_0^{1/2} (\log_5 x + \frac{1}{\ln 5}) dx$.

$$\begin{aligned} & \int_0^{1/2} \log_5 x + \frac{2x}{x \ln 5} dx \\ &= \int_0^{1/2} 2 \log_5 x + \frac{2}{\ln 5} dx \\ &= 2 \left(\log_5 x + \frac{x}{\ln 5} \right) \Big|_0^{1/2} \\ &= 2 \left(\log_5 \frac{1}{2} + \frac{1/2}{\ln 5} \right) - \left[\log_5 0 + \frac{0}{\ln 5} \right] \end{aligned}$$

as $\log_5(0)$ is undefined

Question 31 (4 marks)

Ms Alrubai and Mrs Norman are employed as doctors. They both start on the same annual salary. However, Ms Alrubai negotiated an annual salary increase of \$5 000, while Mrs Norman negotiated an annual salary increase of 7%.

Find their starting annual salary given that after 16 years they have both made the same amount of money in total.

Alrubai - AP	Norman - GP
$S_n = \frac{n}{2} [2a + (n-1)d]$	$S_n = \frac{a(1.07^n - 1)}{1.07 - 1}$
$S_{16} = \frac{16}{2} [2a + (16-1)5000]$	$S_{16} = \frac{a(1.07^{16} - 1)}{0.07}$
$= 8 [2a + 75000]$	$= \frac{a(1.07^{16} - 1)}{0.07}$
$= 16a + 600\,000$ ①	$= \left(\frac{1.07^{16} - 1}{0.07} \right) a$ ①

** Most students used term formula instead of sum $\therefore -2$ as it makes it easier*

$$\therefore 16a + 600\,000 = \left(\frac{1.07^{16} - 1}{0.07} \right) a$$

$$16a - \left(\frac{1.07^{16} - 1}{0.07} \right) a = -600\,000$$

$$\left(16 - \left(\frac{1.07^{16} - 1}{0.07} \right) \right) a = -600\,000$$

** If term formula used incorrectly, -1.*

$$a = \frac{-600\,000}{\left(16 - \left(\frac{1.07^{16} - 1}{0.07} \right) \right)} = \$50\,470.84$$

Question 32 (6 marks)

If Mr Ferrarin wakes up late, the probability that he exercises is $\frac{1}{3}$, and if he wakes up on time the probability that he exercises is $\frac{4}{5}$.

- (a) In a particular week, Mr Ferrarin wakes up on time on Thursday morning and late on Friday morning.

Find the probability that Mr Ferrarin exercised on at least one of these days.

$$1 - P(\text{doesn't work out})$$

$$= 1 - \left(\frac{1}{5} \times \frac{2}{3} \right)$$

$$= \frac{13}{15}$$

- (b) The probability that Mr Ferrarin wakes up on time on a Monday is $\frac{1}{4}$.

- (i) Find the probability that Mr Ferrarin exercises on any given Monday.

$$\frac{1}{4} \times \frac{4}{5} + \frac{3}{4} \times \frac{1}{3}$$

$$= 0.45$$

- (ii) Hence, or otherwise, find the probability that Mr Ferrarin woke up on time on a Monday, given that he exercised that day.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

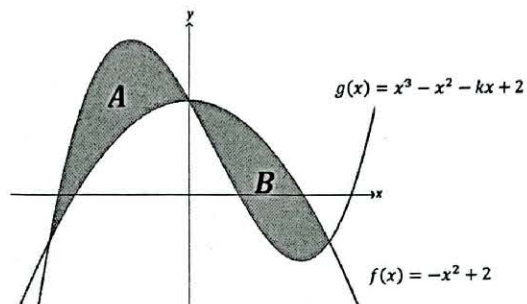
$$= \frac{\frac{1}{4} \times \frac{4}{5}}{0.45} = \frac{4}{9}$$

-27- Problem with these questions:
If one mistake is made,
hard to do CF6.
Most mistakes made
question easier $\therefore$ can't
get $\frac{1}{2}$

Question 33 (4 marks)

The graphs of $f(x) = -x^2 + 2$ and $g(x) = x^3 - x^2 - kx + 2$, where k is a constant, are shown below.

The graphs intersect and create two closed regions, A and B.



Show that these two regions have the same area.

pts of intersection $\rightarrow x^3 - x^2 - kx + 2 = -x^2 + 2$

$$x^3 - kx = 0$$

$$x(x^2 - k) = 0$$

$$x = 0 \text{ or } x^2 - k = 0 \quad \textcircled{1}$$

$$x = \pm \sqrt{k}$$

For A $\rightarrow (x^3 - x^2 - kx + 2) - (-x^2 + 2)$ For B $\rightarrow (-x^2 + 2) - (x^3 - x^2 - kx + 2)$

$$= x^3 - kx$$

$$A = \int_{-\sqrt{k}}^0 x^3 - kx \, dx$$

$$= \left[\frac{x^4}{4} - \frac{kx^2}{2} \right]_{-\sqrt{k}}^0$$

$$= (0) - \left(\frac{(-\sqrt{k})^4}{4} - \frac{k(\sqrt{k})^2}{2} \right)$$

$$= -\frac{k^2}{4} + \frac{k^2}{2}$$

$$= +\frac{k^2}{4} \quad \textcircled{1}$$

$$B = \int_0^{\sqrt{k}} x^3 + kx \, dx$$

$$= \left[\frac{x^4}{4} + \frac{kx^2}{2} \right]_0^{\sqrt{k}}$$

$$= \left(\frac{(\sqrt{k})^4}{4} + \frac{k(\sqrt{k})^2}{2} \right) - (0)$$

$$= \frac{k^2}{4} + \frac{k^2}{2}$$

$$= +\frac{k^2}{4} \quad \textcircled{1}$$

$$\therefore A = B \quad \textcircled{1}$$

* If indefinite integral then -1 because C, needs to be added.

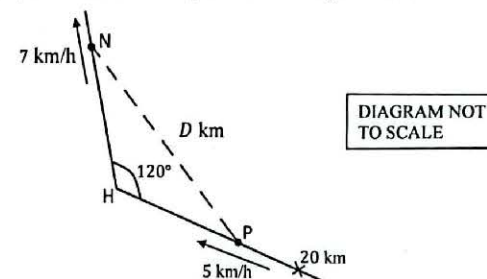
* Most students over complicated question.

* If incorrect intersection is found then CFE.

* If assumption is made about add functions without

Question 34 (5 marks)

Ned Kelly, N, left his hideout, H, and walked along a straight path at 7 km/h. At the same time that Ned left his hideout, the Police, P, are 20 km away from his hideout, walking towards it along a different straight path at 5 km/h. The two paths meet at an angle of 120° .



The distance between Ned Kelly and the police at time t hours is D km.

(a) Show that $D^2 = 39t^2 - 60t + 400$.

$$P = 20 - 5t \quad N = 7t \quad \text{* Done poorly.}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$D^2 = (20 - 5t)^2 + (7t)^2 - 2(20 - 5t)(7t) \cos(120^\circ) \quad \textcircled{1}$$

$$= 400 - 200t + 25t^2 + 49t^2 - (280t - 70t^2) \left(-\frac{1}{2}\right)$$

$$= 400 - 200t + 74t^2 + 140t - 35t^2$$

$$= 39t^2 - 60t + 400 \quad \textcircled{1}$$

(b) Find the minimum distance, to the nearest kilometre, between the police and Ned Kelly.

D is a minimum when D^2 is.

$\therefore$ Find min st pt for D^2

$$D^2 = 39\left(\frac{10}{13}\right)^2 - 60\left(\frac{10}{13}\right) + 400$$

$$= 371.5976331$$

$$\frac{dD^2}{dt} = 78t - 60$$

$$\text{st pt when } \frac{dD^2}{dt} = 0$$

$$0 = 78t - 60$$

$$t = \frac{60}{78} = \frac{10}{13} \quad \textcircled{1}$$

$$D = 19.28 \text{ km (2dp)}$$

$$\therefore \text{min st pt.}$$

$$\frac{dD^2}{dt} = 78t - 60$$

$$\frac{dD^2}{dt} = 78t - 60$$

$$\frac{dD^2}{dt} = 78t - 60$$

$$\frac{dD^2}{dt} = 78t - 60$$

$$\frac{dD^2}{dt} = 78t - 60$$

* Most students forget to test to show minimum OR use property of original function (concave up parabola).